

**FOR ODD PRIMES $p > q$ WITH $q \mid p^2 - 1$, EVERY
NONASSOCIATIVE RIGHT BOL LOOP OF ORDER pq HAS
RIGHT MULTIPLICATION GROUP OF ORDER p^2q**

ABSTRACT. Kinyon, Nagy and Vojtěchovský classified the right Bol loops of order pq for odd primes $p > q$ with $q \mid p^2 - 1$, and proved that the right multiplication group of such a loop has order p^kq with $k \in \{2, 3\}$. Their Problem 7.1 asks whether $k = 2$ always. Granting their classification, we answer yes: the p^3q branch is empty. Once the right translations are put in affine form the proof is short linear algebra, and it identifies the module: for each such pair p, q and every nonassociative right Bol loop Q of order pq , $\text{Mlt}_r(Q) \cong \mathbb{F}_p^2 \rtimes_C \mathbb{Z}_q$, where C is the companion matrix of $X^2 - \lambda X + 1$ over $\mathbb{F}_p$, with $\lambda = w + w^{-1}$ for a primitive q -th root of unity w , and $\text{RIInn}(Q) \cong \mathbb{Z}_p$. This extends their Theorem 1.1(v) from the Bruck loop $B_{p,q}$ to the whole family.

1. THE PROBLEM

A loop Q is a *right Bol loop* if $((zx)y)x = z((xy)x)$ for all $x, y, z \in Q$. Its *right multiplication group* is $\text{Mlt}_r(Q) = \langle R_u : u \in Q \rangle \leq \text{Sym}(Q)$, where $R_u : x \mapsto xu$, and its *right inner mapping group* $\text{RIInn}(Q)$ is the stabiliser of the identity in $\text{Mlt}_r(Q)$, so that $|\text{Mlt}_r(Q)| = |Q| \cdot |\text{RIInn}(Q)|$.

Let $p > q$ be odd primes with $q \mid p^2 - 1$. Kinyon, Nagy and Vojtěchovský [1] classify the right Bol loops of order pq (their Theorem 1.1(ii)–(iii)), show in Theorem 1.1(iv) that $|\text{Mlt}_r(Q)| = p^kq$ with $k \in \{2, 3\}$ for every nonassociative such loop, and prove in Theorem 1.1(v) that $\text{Mlt}_r(B_{p,q}) \cong (\mathbb{Z}_p \times \mathbb{Z}_p) \rtimes \mathbb{Z}_q$ for the unique nonassociative right *Bruck* loop $B_{p,q}$ of that order. The open half of the dichotomy is the last item of their Section 7:

Problem 7.1 (Kinyon, Nagy, Vojtěchovský). Let $p > q$ be odd primes such that q divides $p^2 - 1$, and let Q be a nonassociative right Bol loop of order pq . Is the order of the right multiplication group of Q equal to p^2q ?

The problem is quoted from the arXiv e-print [arXiv:1611.05785v3](https://arxiv.org/abs/1611.05785v3) of [1], whose numbering we follow throughout.

Theorem 1. *Let $p > q$ be odd primes with $q \mid p^2 - 1$ and let Q be a nonassociative right Bol loop of order pq . Then*

$$|\text{Mlt}_r(Q)| = p^2q,$$

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so the $k = 3$ branch of [1, Theorem 1.1(iv)] is empty. Consequently $\text{RInn}(Q) \cong \mathbb{Z}_p$.

The proof is in Section 3. It uses [1, Theorem 1.1(ii)–(iii)], which supplies the normal form of Section 2 and is not reproved here, and it uses neither Theorem 1.1(iv) nor Theorem 1.1(v).

2. THE OBJECT

Since $q \mid p^2 - 1$ and q is an odd prime, $p \equiv \pm 1 \pmod{q}$ and exactly one of $q \mid p - 1$, $q \mid p + 1$ holds. Fix a primitive q -th root of unity $w \in \mathbb{F}_{p^2}$; then $w \in \mathbb{F}_p$ iff $q \mid p - 1$, and $w^p = w^{-1}$ iff $q \mid p + 1$. For $\gamma \in \mathbb{F}_{p^2}$ put

$$(1) \quad u_i := \gamma w^i + (1 - \gamma)w^{-i} \quad (i \in \mathbb{Z}_q),$$

so $u_0 = 1$, and let

$$\Gamma := \{\gamma \neq 0 : u_i \in \mathbb{F}_p \text{ and } u_i \neq 0 \text{ for all } i \in \mathbb{Z}_q\} / (\gamma \sim 1 - \gamma).$$

The identification $\gamma \mapsto 1 - \gamma$ replaces u by $(u_{-i})_i$ and so does not change the loop below. Concretely Γ is $\mathbb{F}_p^*$ cut down by the condition $1 - \gamma^{-1} \notin \langle w \rangle$ when $q \mid p - 1$, and the trace-one line $\{\gamma : \gamma + \gamma^p = 1\}$ cut down by the same condition when $q \mid p + 1$.

For $\gamma \in \Gamma$ let Q_γ be the loop on $\mathbb{Z}_q \times \mathbb{F}_p$ with

$$(2) \quad (i, j)(k, l) = \left(i + k, m + (j + m) \frac{u_i}{u_{i+k}} \right), \quad m := l \frac{u_k}{u_k + 1},$$

neutral element $(0, 0)$. This is [1, Theorem 1.1(iii)] written with $\theta_i = u_i^{-1}$: their displayed product is $(i, j)(k, l) = (i + k, l(1 + \theta_k)^{-1} + (j + l(1 + \theta_k)^{-1})\theta_i^{-1}\theta_{i+k})$, and substituting $\theta = u^{-1}$ gives (2). Their Theorem 1.1(ii)–(iii) states that every right Bol loop of order pq is isomorphic to $\mathbb{Z}_{pq}$ or to exactly one Q_γ ; Q_γ is associative iff $\gamma = 1$, which occurs only when $q \mid p - 1$ and gives the nonabelian group of order pq ; and $\gamma = 1/2$ gives the Bruck loop $B_{p,q}$.

Everything below depends on Q_γ only through the vector $u = (u_0, \dots, u_{q-1}) \in \mathbb{F}_p^q$.

3. THE PROOF

Throughout, $\gamma \in \Gamma$ with $\gamma \neq 1$, u is given by (1), $Q = Q_\gamma$, and S denotes the cyclic shift $(Sr)_i = r_{i+1}$ on $\mathbb{F}_p^{\mathbb{Z}_q}$ and on $\mathbb{F}_{p^2}^{\mathbb{Z}_q}$.

Step 1: the right translations are affine. By (2), for $(k, l) \in Q$,

$$(3) \quad R_{(k,l)} : (i, j) \mapsto (i + k, \alpha_i j + \beta_i), \quad \alpha_i = \frac{u_i}{u_{i+k}}, \quad \beta_i = m(1 + \alpha_i),$$

with $m = l u_k / (u_k + 1)$ as in (2). All of u_k , $u_k + 1 = u_k + u_0$ and u_i are nonzero, so $\alpha_i \in \mathbb{F}_p^*$ and m is defined; as l runs over $\mathbb{F}_p$ so does m . Hence $\text{Mlt}_r(Q)$ sits inside the group A of maps $(i, j) \mapsto (i + c, a_i j + b_i)$ with

$a \in (\mathbb{F}_p^*)^{\mathbb{Z}_q}$, $b \in \mathbb{F}_p^{\mathbb{Z}_q}$, $c \in \mathbb{Z}_q$, whose product is $(a, b, c)(a', b', c') = (a'', b'', c+c')$ with $a''_i = a_i a'_{i+c}$. $\square$

Step 2: the kernel is exactly the pure translations, and $|\text{Mlt}_r(Q)| = p^{\dim P} q$. Let $\psi: A \rightarrow (\mathbb{F}_p^*)^{\mathbb{Z}_q} \rtimes \mathbb{Z}_q$ kill the translation part b . By (3), $\psi(R_{(k,l)}) = ((u_i/u_{i+k})_i, k)$ independently of l . Writing $h := \psi(R_{(1,0)})$ and using the product rule of Step 1, $h^k = ((u_i/u_{i+k})_i, k)$ by telescoping, so $\psi(\text{Mlt}_r(Q)) = \langle h \rangle$; and $h^k \neq 1$ for $0 < k < q$ because its $\mathbb{Z}_q$ -part is k , while $h^q = 1$. Thus $\langle h \rangle$ has order exactly q and the projection $\langle h \rangle \rightarrow \mathbb{Z}_q$ is an isomorphism. Therefore an element of $\text{Mlt}_r(Q)$ with trivial $\mathbb{Z}_q$ -part already has trivial linear part, i.e.

$$P := \ker(\psi|_{\text{Mlt}_r(Q)}) = \{ (i, j) \mapsto (i, j + t_i) \} \cap \text{Mlt}_r(Q) \leq \mathbb{F}_p^{\mathbb{Z}_q},$$

an elementary abelian p -group, and

$$(4) \quad |\text{Mlt}_r(Q)| = |P| \cdot q = p^{\dim_{\mathbb{F}_p} P} q.$$

$\square$

Step 3: P is a span of shifts, after one rescaling. Put $s := R_{(1,0)}$; by (3) it has $\beta \equiv 0$, and $s^k = R_{(k,0)}$. For general (k, l) ,

$$R_{(k,l)} = n_{k,l} s^k, \quad n_{k,l} \in \mathbb{F}_p^{\mathbb{Z}_q} \text{ the translation by } t_i = m \left(1 + \frac{u_{i+k}}{u_i} \right).$$

Since $\text{Mlt}_r(Q)$ is generated by the $R_{(k,l)}$ and $s^q = 1$, the group P is generated by the s -conjugates of the $n_{k,l}$; being elementary abelian, it is their $\mathbb{F}_p$ -span. Conjugation by s sends a translation vector t to $(t_{i+1}u_{i+1}/u_i)_i$. Every u_i is invertible, so $r_i := t_i u_i$ is a change of coordinates on $\mathbb{F}_p^{\mathbb{Z}_q}$, and in it that conjugation is the plain shift S . In the same coordinates

$$r_i = t_i u_i = m(u_i + u_{i+k}) = m x_i^{(k)}, \quad x_i^{(k)} := u_i + u_{i+k}.$$

As l runs over $\mathbb{F}_p$, m runs over $\mathbb{F}_p$. Hence, in r -coordinates,

$$(5) \quad P = R := \text{span}_{\mathbb{F}_p} \{ S^a x^{(k)} : a, k \in \mathbb{Z}_q \}, \quad \text{so} \quad \dim P = \dim R.$$

$\square$

Step 4: every generator lies in one fixed plane, so $\dim R \leq 2$. Set $v_1 := (w^i)_{i \in \mathbb{Z}_q}$ and $v_{-1} := (w^{-i})_{i \in \mathbb{Z}_q}$ in $\mathbb{F}_{p^2}^{\mathbb{Z}_q}$, so that $u = \gamma v_1 + (1 - \gamma)v_{-1}$ by (1) and $Sv_{\pm 1} = w^{\pm 1}v_{\pm 1}$. Then, for every k ,

$$(6) \quad x^{(k)} = \gamma(1 + w^k)v_1 + (1 - \gamma)(1 + w^{-k})v_{-1},$$

because $x_i^{(k)} = u_i + u_{i+k} = \gamma(1 + w^k)w^i + (1 - \gamma)(1 + w^{-k})w^{-i}$. So every generator of R , and hence every S -shift of one, lies in $W := \mathbb{F}_{p^2}v_1 + \mathbb{F}_{p^2}v_{-1}$, which is S -stable and has $\dim_{\mathbb{F}_{p^2}} W = 2$ since $w \neq w^{-1}$. A family of vectors of $\mathbb{F}_p^{\mathbb{Z}_q}$ that is $\mathbb{F}_p$ -independent stays independent over $\mathbb{F}_{p^2}$, so

$$\dim_{\mathbb{F}_p} R \leq \dim_{\mathbb{F}_{p^2}} W = 2.$$

□

Step 5: two of them are already independent, so $\dim R = 2$. Take $k = 0$: $x^{(0)} = 2u = 2\gamma v_1 + 2(1 - \gamma)v_{-1}$ and $Sx^{(0)} = 2\gamma w v_1 + 2(1 - \gamma)w^{-1}v_{-1}$. In the basis (v_1, v_{-1}) of W their determinant is

$$(7) \quad \det \begin{pmatrix} 2\gamma & 2(1 - \gamma) \\ 2\gamma w & 2(1 - \gamma)w^{-1} \end{pmatrix} = 4\gamma(1 - \gamma)(w^{-1} - w).$$

Every factor is nonzero: $4 \neq 0$ because p is odd; $w^{-1} \neq w$ because w has odd order $q > 1$; $\gamma \neq 0$ by the definition of Γ ; and $1 - \gamma \neq 0$ because $\gamma = 1$ is the associative member, excluded by hypothesis. So $x^{(0)}$ and $Sx^{(0)}$ are independent over $\mathbb{F}_{p^2}$, a fortiori over $\mathbb{F}_p$, and $\dim R \geq 2$. With Step 4, $\dim R = 2$; with (5) and (4),

$$|\text{Mlt}_r(Q)| = p^2q.$$

Finally $|\text{RInn}(Q)| = |\text{Mlt}_r(Q)|/|Q| = p$, so $\text{RInn}(Q) \cong \mathbb{Z}_p$. This proves Theorem 1. □

Corollary 2. *$R = W \cap \mathbb{F}_p^{\mathbb{Z}_q}$, independently of γ , and the shift S acts on the 2-dimensional $\mathbb{F}_p$ -space R with characteristic polynomial $X^2 - \lambda X + 1$, where $\lambda := w + w^{-1} \in \mathbb{F}_p$, so with eigenvalues w and w^{-1} . Hence, for a fixed admissible pair p, q , all nonassociative right Bol loops of order pq have mutually isomorphic right multiplication groups,*

$$\text{Mlt}_r(Q) \cong \mathbb{F}_p^2 \rtimes_C \mathbb{Z}_q, \quad C = \begin{pmatrix} 0 & -1 \\ 1 & \lambda \end{pmatrix} \in \text{GL}_2(\mathbb{F}_p),$$

the generator of $\mathbb{Z}_q$ acting on $\mathbb{F}_p^2$ by the companion matrix C of $X^2 - \lambda X + 1$. So [1, Theorem 1.1(v)] extends from $B_{p,q}$ to the whole family.

Proof. $R \subseteq W \cap \mathbb{F}_p^{\mathbb{Z}_q}$ by Step 4, and $\dim_{\mathbb{F}_p}(W \cap \mathbb{F}_p^{\mathbb{Z}_q}) = 2$ in both branches. If $q \mid p - 1$ then $w \in \mathbb{F}_p$, so $v_{\pm 1}$ themselves lie in $\mathbb{F}_p^{\mathbb{Z}_q}$ and span a 2-dimensional $\mathbb{F}_p$ -subspace of $W \cap \mathbb{F}_p^{\mathbb{Z}_q}$, which by Step 4 cannot be larger. If $q \mid p + 1$ then $w^p = w^{-1}$, so the Frobenius σ of $\mathbb{F}_{p^2}/\mathbb{F}_p$ interchanges v_1 and v_{-1} coordinatewise; thus W is σ -stable and Galois descent gives $\dim_{\mathbb{F}_p}(W \cap \mathbb{F}_p^{\mathbb{Z}_q}) = \dim_{\mathbb{F}_{p^2}} W = 2$. Either way R and $W \cap \mathbb{F}_p^{\mathbb{Z}_q}$ are 2-dimensional with the first inside the second, so they are equal — and W does not involve γ . Since $Sv_{\pm 1} = w^{\pm 1}v_{\pm 1}$, the shift acts on $R \otimes \mathbb{F}_{p^2} = W$ with eigenvalues w, w^{-1} ; as R is an $\mathbb{F}_p$ -space its characteristic polynomial lies in $\mathbb{F}_p[X]$, so it is $(X - w)(X - w^{-1}) = X^2 - \lambda X + 1$ with $\lambda = w + w^{-1} \in \mathbb{F}_p$. The two eigenvalues are distinct, so the matrix of S on R in any $\mathbb{F}_p$ -basis is conjugate in $\text{GL}_2(\mathbb{F}_p)$ to the companion matrix C of that polynomial. By Step 2 the extension $1 \rightarrow P \rightarrow \text{Mlt}_r(Q) \rightarrow \mathbb{Z}_q \rightarrow 1$ splits through s , with $P \cong R$ as an $\mathbb{F}_p[\mathbb{Z}_q]$ -module, so $\text{Mlt}_r(Q)$ is determined up to isomorphism by that module, which is now the same for every γ and is $\mathbb{F}_p^2$ with the generator of $\mathbb{Z}_q$ acting by C . □

4. SCOPE OF THE ACCOMPANYING PROGRAM

The proof above covers all admissible (p, q) at once and is what the theorem rests on. The accompanying program `verify.py` checks it rather than replaces it, in exact integer arithmetic in $\mathbb{F}_p$ and $\mathbb{F}_{p^2}$; its transcript is `verify.output.txt`. On the cells with $pq \leq 105$ it builds the multiplication tables from (2) and from KNV's θ -form independently, checks that the two agree entrywise, and verifies the loop axioms and the right Bol identity for every member. It checks Steps 1–5 as identities over the 29 admissible cells with $pq \leq 255$, some of them only over the cells with $pq \leq 105$. Over those 29 cells, all 490 members that the imported classification marks nonassociative have $|\text{Mlt}_r| = p^2q$ by brute-force closure of the right translations, using nothing from the proof; associativity itself is retested only on the cells with $pq \leq 105$ and on individual samples. A dimension sweep reaches $p \leq 400$ for $q \leq 13$. The program assumes [1, Theorem 1.1(ii)–(iii)], which the theorem therefore also assumes; it examines no loop outside the family (2) and no order other than pq .

No priority is claimed. The vector u of (1) obeys the recurrence $u_{i+2} = \lambda u_{i+1} - u_i$ that [1] attribute to Niederreiter and Robinson, and Step 4 is that recurrence in the form of the shift-invariant plane it determines.

REFERENCES

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